

Problematic Internet Usage in Youths: An Ethical Audit of Bias

Ashley Tewari

Center for Data Science
NYU

Luke Sullivan

Center for Data Science
NYU

Abstract

The internet has fundamentally changed how people interact with one another and the world. Automated decision systems, such as ones created for the Child Mind Institute’s Problematic Internet Use Kaggle competition, explore potential predictions for diagnoses of certain disorders. In this paper we will conduct an audit on the 1 model from the competition to determine its performance capabilities. We will later discuss these findings and what they indicate moving forward.

1 Introduction

The effect of the digital world on children’s well-being has become a hot topic in the medical and mental health sectors. Growing concern over harmful internet use on this vulnerable population has prompted a wave of diagnostic measures.

Diagnostic models that predict mental health and behavioral disorders are ethically paramount, making them a prevalent case study for algorithmic fairness and accountability. To explore this, we propose an audit of an ADS developed for the *Child Mind Institute, Problematic Internet Use* Kaggle competition. The purpose of this ADS is to predict a youth’s Severity Impairment Index (*sii*) by relying on physical proxy data, mainly physical activity metrics and internet usage behavior.

By auditing this system, we aim to uncover potential performance disparities across demographic subpopulations (such as age and sex) and evaluate the ethical trade-offs of misclassifying a youth’s mental health severity based on these physical proxies.

We will audit the 1st-place winning implementation made by Lennart Haupts. This solution utilizes a voted ensemble of regression models (LGBM, XGBoost, CatBoost, and ExtraTrees) to predict the continuous *PCIAT* – *PCIAT_Total* score, which is then converted into the final *sii* classification.

The pipeline has extensive data processing, including Lasso imputation for missing tabular data, PCA dimensionality reduction, and explicit feature normalization.

2 Input and Output

2.1 Input

2.1.1 Data Collection & Source

The data used by this ADS originates from the Healthy Brain Network (HBN), a large-scale initiative by the Child Mind Institute to collect biomedical data on 5,000+ children aged 5–21 in the New York City area. Participants were recruited through community referrals targeting families who had concerns about their child’s learning, attention, or psychiatric health, meaning the dataset is a clinically-referred, not population-representative, sample. Data collection occurred across multiple sessions and included clinical interviews, cognitive assessments, physical fitness evaluations, and multi-day wrist actigraphy. The specific data provided through the Kaggle competition is a pre-processed subset of the HBN, released in tabular (`train.csv/test.csv`) and time-series parquet form.

A critical structural feature of this data is that the ground-truth label, the Severity Impairment Index (*sii*), is derived from the Parent-Child Internet Addiction Test (PCIAT), which was administered only to a subset of participants. Rows without a *sii* value are dropped from training entirely, reducing the effective training set and introducing a potential selection bias — participants for whom PCIAT data exists may differ systematically from those for whom it does not.

2.1.2 Input Features

Basic Demographics Two features: `Basic_Demos-Age` (continuous float, years) and `Basic_Demos-Sex` (binary integer, 0/1). Age

spans approximately 5–22 years across the training set. Sex is a protected attribute and is directly used as a model input, which raises fairness considerations.

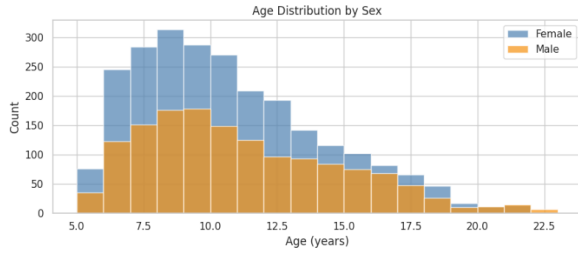


Figure 1: Age distribution by sex across the training set.

Children’s Global Assessment Scale A single numeric score (CGAS-CGAS_Score) representing a clinician-assigned global functioning rating (1–100 scale). Contains a seasonal enrollment indicator (CGAS-Season), dropped during feature engineering.

Physical Measurements Continuous measurements including height, weight, BMI, systolic/diastolic blood pressure, and resting heart rate. These are collected via direct clinical assessment. Physical-BMI and the BIA-derived BIA-BMI are both present and partially redundant; the model constructs a normalized composite (BMI_mean_norm) corrected for age-group-specific population means.

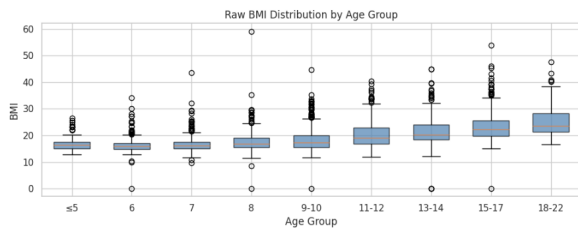


Figure 2: Raw BMI distribution by age group. Median BMI increases monotonically with age, consistent with expected pediatric growth patterns.

Fitness Endurance Captures a timed endurance test (duration in seconds, Fitness_Endurance-Time_Sec, and associated maximum heart rate). Exhibits the highest missingness of any tabular feature group, with Fitness_Endurance-Time_Sec and related columns exceeding 80% missing in the training set, reflecting that the timed endurance test was completed by only a small subset of participants.

Fitness Gram Challenge A battery of youth fitness assessments: grip strength dominant/non-dominant hand (FGC_GSD/FGC_GSND), curl-ups (FGC_CU), push-ups (FGC_PU), trunk lift (FGC_TL), and bilateral sit-and-reach (FGC_SRL/FGC_SRR). Each measure has a corresponding ordinal Healthy Fitness Zone indicator (*_Zone). Grip values are hard-clipped to [9, 60] kg during cleaning; all zone columns are dropped after engineering age-normalized aggregate features (FGC_Zones_mean/min/max, GS_max/min). Missing rates for FGC features are moderate to high.

Bioelectrical Impedance Analysis The largest single feature group, covering body composition outputs from a bioimpedance device: fat percent-age, fat-free mass (FFM), fat mass index (FMI), fat-free mass index (FFMI), basal metabolic rate (BMR), daily energy expenditure (DEE), bone mineral content (BMC), skeletal muscle mass (SMM), total body water (TBW), extracellular water (ECW), intracellular water (ICW), lean dry mass (LDM), and lean soft tissue (LST). All are continuous floats. Several exhibit implausible outliers that are nullified during cleaning (e.g., BMR > 4000 kcal, fat percentage outside [5, 60]%). The model engineers age-normalized variants (BMR_norm, DEE_norm, FFM_norm) and the ratio ICW_ECW. BIA features have among the highest missing rates of the tabular feature groups, reflecting device availability constraints at collection sites.

Physical Activity Questionnaire Self-reported activity level summary scores from either the adult proxy (PAQ_A) or child-completed (PAQ_C) versions of the Physical Activity Questionnaire. Only PAQ_A_Total is retained after feature engineering. These are ordinal-valued summary scales.

Sleep Disturbance Scale A single summary score from the Sleep Disturbance Scale for Children. Continuous integer scale.

Pre-Intervention Educational History A single feature capturing computer usage context, specifically computer use during school hours or at home (integer ordinal).

Missing Value Summary Missing data is pervasive in this dataset. Features with >30% missingness include most BIA columns and Fitness Endurance metrics. The model addresses this via a Lasso-based predictive imputer (LassoCV with 5-

fold CV): for features with less than 40% missing, values are imputed using predictions from a Lasso model trained on other features; for features exceeding this threshold, mean imputation is applied. This introduces a key auditability concern — imputed values are not flagged as synthetic, meaning downstream fairness analyses cannot distinguish observed from imputed inputs.

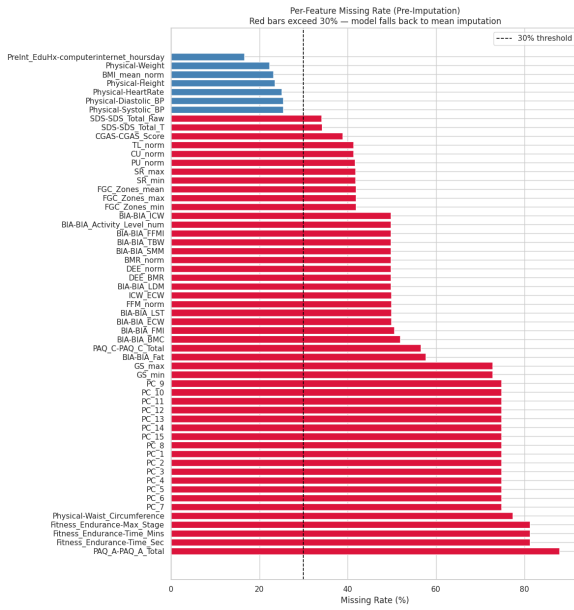


Figure 3: Per-feature missing rate (pre-imputation). Red bars exceed the 30% threshold at which the model falls back to mean imputation.

Actigraphy Time-Series Features A subset of participants wore wrist accelerometers that collected high-frequency sensor data stored as partitioned Parquet files. The raw time-series contains five channels per timestep: Euclidean Norm Minus One (enmo, a gravity-corrected movement magnitude), wrist angle (anglez), ambient light (light), battery voltage (battery_voltage), and a binary non-wear flag (non-wear_flag). Rather than training sequence models directly on the raw series, the implementation collapses the time-series into 74 summary statistics per participant via the `time_features()` function. For each of the four sensor channels, six statistics (mean, std, max, min, first-order difference mean, first-order difference std) are computed under three temporal masks: unrestricted, nighttime (22:00–05:00), and daytime (07:00–20:00), yielding $4 \times 3 \times 6 = 72$ features, plus the mean non-wear rate and active-ENMO sum (≥ 0.05 g). These 74 features are then standardized and reduced to 15 principal components

via PCA. As shown in Figure 5, PC_1 and PC_2 alone account for approximately 22% and 12% of variance respectively, with all 15 components collectively capturing approximately 82% of total actigraphy variance while controlling dimensionality. Not all participants have actigraphy data; for those without it, the PCA components are left-joined as NaN and subsequently imputed.

2.2 Output

2.2.1 Output Variable

The SII The ADS’ output is the Severity Impairment Index (*sii*). This results in an integer with a value from 0 to 3 that indicates the level of severity for an individual’s problematic internet usage.

- **0 — None:** PCIAT Total < 30
- **1 — Mild:** $30 \leq \text{PCIAT Total} < 50$
- **2 — Moderate:** $50 \leq \text{PCIAT Total} < 80$
- **3 — Severe:** PCIAT Total ≥ 80

This model does not expressly predict the value of the *sii*, but instead predicts the score of the Parent-Child Internet Addiction Test. This value is then converted to an ordinal one that fits the designated pre-determined thresholds.

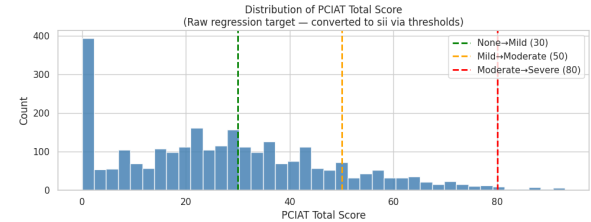


Figure 4: The distribution of PCIAT score. The dashed lines show the thresholds that are used to convert the score to *sii*.

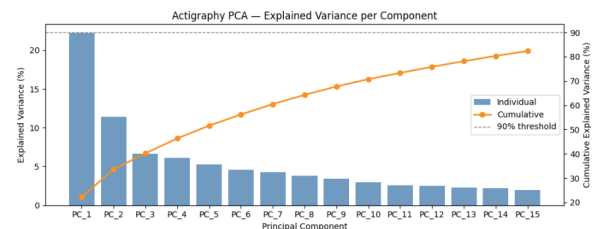


Figure 5: Explained variance per actigraphy PCA component, with cumulative variance overlay. The 15 components collectively capture approximately 82% of total variance.

Class Distribution There is great imbalance within the training dataset with a large emphasis on classification 0 and 1 for the *sii*. The classifications of 2 and 3 are underrepresented as the majority of participants fall into the lower severity categories. This is a risk for the accuracy of the model as it makes the model more likely to be biased toward predicting individuals as 0 or 1. We can see in Figure 6 that the distribution is extremely skewed toward the None and Mild classifications. It's essential that this is addressed so that model does not persist as biased in this area.

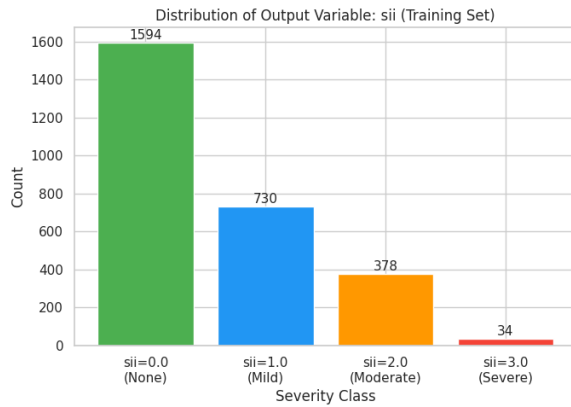


Figure 6: The distribution of the *sii*.

Evaluation Metric The Quadratic Weighted Cohen's Kappa score indicates how our model is performing on a scale of 1 to -1. This metric takes into consideration the ordinal nature of the *sii*. This means that there is a larger penalty for predictions that are further from the true value. This is in contrast to penalizing predictions based on simple accuracy, which would mean all off-predictions would have the same penalty. If a model receives This model has a 0.47 for the Quadratic Weighted Cohen's score, indicating that there is moderate-leaning agreement between the predicted values and the true values.

- **1.0** — Perfect agreement with true
- **0.6 – 1.0** — Substantial agreement
- **0.4 – 0.6** — Moderate agreement
- **0.2 – 0.4** — Fair agreement
- **0.0** — No better than random chance
- **< 0.0** — Worse than random chance

Feature Correlation As shown in Figure 7, the group-average correlation heatmap reveals that BIA features have a mean absolute intra-group correlation of approximately 0.48, reflecting their shared physiological derivation from the same bioimpedance device. BIA features also show moderate correlation (~ 0.37) with Physical measurements such as height and weight, which is expected given that body composition is partly a function of size. By contrast, the actigraphy principal components and questionnaire-based features (PAQ, SDS) show near-zero correlation with all other groups, suggesting they contribute independent signal. This high intra-BIA correlation warrants caution when interpreting tree-based feature importances, as importance may be fragmented arbitrarily across structurally redundant features.

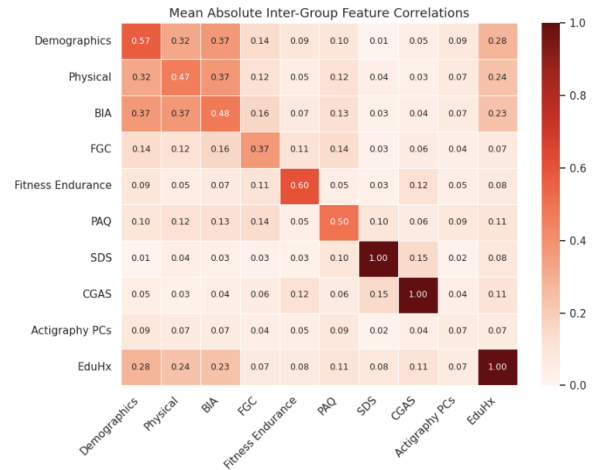


Figure 7: Mean absolute inter-group feature correlations (pre-imputation). BIA features exhibit the highest intra-group correlation (0.48), while actigraphy PCs and questionnaire features are largely independent of all other groups.

3 Implementation and Validation

- **Data Cleaning and Preprocessing:** We will describe the dataset's variable attributes and indicate the steps we took to clean the data.

Data Cleaning and Preprocessing As shown in Figure 7... For data accuracy, baseline thresholds were implemented for various metrics. BMR, daily energy expenditure, body fat percentage, bone mineral content, and grip strength values that were unrealistic were nullified.

- **Feature Engineering:** We will describe the

engineered features we have for the raw inputs. These include age-normalized BMI, zone aggregates, and the actigraphy summary statistics.

- **Model Implementation:** We will describe the ensemble model, and explain how the models and predictions are trained and produced.
- **Validation:** We will describe how the 1 model we picked from the competition is evaluated.

4 Outcomes

- **Accuracy Analysis:** We will determine the accuracy of the chosen model for different demographics. We will implement the Quadratic Weighted Cohen's Kappa scoring system as our metric.
- **Fairness Analysis:** We will determine fairness of the model for the protected demographics of age and group. We will determine whether the model underestimates or overestimates severity of different demographics.
- **Robustness and Stability:** We will determine the robustness of the model by looking at how sensitive it is to missing values. For example, we will look at whether or not performance of the model lowers when there are missing values. We will also cross-examine the model with different random seeds.

5 Summary

- **Data Appropriateness:** We will describe the fit of the proxy data used by the model for predicting problematic internet usage. We will also discuss limitations that come from the nature of the sample.
- **Robustness, Accuracy, and Fairness:** We will summarize our insights on how robust, accurate, and fair the model is and indicate the stakeholders.
- **Deployment Considerations:** We will discuss whether the chosen model is appropriate for use in a clinical setting and assess the potential risks involved if so.
- **Recommendations:** We will include our recommendations for how the model can be corrected and improved from data collection to model implementation.